EE1-10A MATHEMATICS I

1. a) i) $(1-i)^3 = \left(\sqrt{(}2)e^{-\pi i/4}\right)^3 = 2^{3/2}e^{-3\pi i/4} = -2-2i$ [1]

ii) $\dfrac{1-i}{1+i}\dfrac{(1-i)}{(1-i)} = \dfrac{-2i}{2} = -i$ [1]

iii) $\left(\dfrac{1+\sqrt{3}i}{2}\right)^{10} = \left(e^{\pi i/3}\right)^{10} = e^{10\pi i/3} = e^{4\pi i/3} = -\dfrac{1}{2} - \dfrac{\sqrt{3}}{2}i$ [2]

Many people left this as $e^{10\pi i/3}$ *or* $e^{4\pi i/3}$ *or* $\cos(10\pi/3)+i\sin(10\pi/3)$. *Always simplify!*

b) $(2+2i)^{1/3} = \left(\sqrt{8}e^{i(\pi/4+2n\pi)}\right)^{1/3} = \sqrt{2}e^{i(\pi/12+2n\pi/3)} \quad n = 0,1,2$ [2]

$\sqrt{2}e^{\pi i/12}, \sqrt{2}e^{3\pi i/4}, \sqrt{2}e^{17\pi i/12}$ [2]

Many had $8^{1/6}$ *or* $(2\sqrt{2})^{1/3}$. *Simplify, whenever possible.*

c) $\lim\limits_{x\to\pi/6} \dfrac{\cos(3x)}{\tan(2x)-\sqrt{3}} = \text{“}\dfrac{0}{0}\text{”} \Rightarrow l'Hopital$

$= \lim\limits_{x\to\pi/6} \dfrac{-3\sin(3x)}{2\sec^2(2x)} = -\dfrac{3}{2}\sin(\pi/2)\cos^2(\pi/3) = -\dfrac{3}{2}\left(\dfrac{1}{2}\right)^2 = -\dfrac{3}{8}$ [4]

A substantial minority didn't recognize this as appropriate for l'Hopital, and tried various, unsuccessful manipulations, e.g multiplying numerator and denominator by $\cos 2x$.

d) Let $y = x^x$, so $\ln y = x\ln x \to 0$, as $x \to 0$; hence $y \to 1$ and $\lim\limits_{x\to 0} x^x = 1$ [4]

Many people got as far as $\ln y = x\ln x$ *but then didn't know where to go next. Some of these tried obtaining* dy/dx. *Others tried* $y = e^{x\ln x}$ *and then had trouble with the limit.*

e) $\ln y = \ln\left(x^{\ln x}\right) = (\ln x)^2$. Differentiate:

$\dfrac{1}{y}\dfrac{dy}{dx} = 2\ln x\left(\dfrac{1}{x}\right) \Rightarrow \dfrac{dy}{dx} = 2(\ln x)x^{\ln x - 1}$. [4]

Most people who obtained $\ln y = \ln\left(x^{\ln x}\right) = (\ln x)^2$ *were fine from then on, but some tried* $y = e^{\ln(x^{\ln x})}$ *which got complicated with the chain rule.*

f) Let $u = 1-3x^2 \Rightarrow du = -6xdx$ so

$\int \dfrac{2x}{(1-3x^2)^{1/3}}\,dx = -\dfrac{1}{3}\int u^{-1/3}du = -\dfrac{1}{3}\dfrac{u^{2/3}}{2/3} = -\dfrac{1}{2}u^{2/3}$. It's an improper integral, due to the discontinuity at $1/\sqrt{3}$, so split into two:

$$-\frac{1}{2}\left[\lim_{k\to 1/\sqrt{3}^-}\left[(1-3x^2)^{2/3}\right]_0^k + \lim_{k\to 1/\sqrt{3}^+}\left[(1-3x^2)^{2/3}\right]_k^1\right]$$

$$= -\frac{1}{2}\left[\lim_{k\to 1/\sqrt{3}^-}(1-3k^2)^{2/3} - 1 + (-2)^{2/3} - \lim_{k\to 1/\sqrt{3}^+}(1-3k^2)^{2/3}\right]$$

$= -\frac{1}{2}\left(4^{1/3}-1\right)$, as the two terms involving k cancel. [4]

Less than 10 people noticed the discontinuity. The remainder ignored it and with the correct substitution, etc, got the right result, with no penalty imposed.

g) Let $x = \sin u \Rightarrow dx = \cos u\, du$ and integral becomes

$$\int_0^{\pi/2} \sqrt{1-\sin^2 u}\cos u\, du = \int_0^{\pi/2} \cos^2 u\, du$$

$$= \tfrac{1}{2}\int_0^{\pi/2} \cos(2u)+1\, du = \tfrac{1}{2}\left[\tfrac{1}{2}\sin(2u)+u\right]_0^{\pi/2} = \tfrac{\pi}{4} \quad [4]$$

Some people didn't change the limits and then had difficulty with the solution in terms of $\frac{1}{2}\left[\frac{1}{2}\sin(2\sin^{-1}(x))+\sin^{-1}x\right]$.

h) From the definition,

$$Y_n * X_n = \sum_{m=-\infty}^{\infty} Y_{n-m}X_m$$

We now make a substitution: $m = n-r$. For any fixed n this is a 1-1 mapping between m and r with the range $-\infty \le m \le \infty$ mapping to $\infty \le r \le -\infty$. Therefore

$$\begin{aligned} Y_n * X_n &= \sum_{m=-\infty}^{\infty} Y_{n-m}X_m \\ &= \sum_{r=\infty}^{-\infty} Y_r X_{n-r} \\ &= \sum_{r=-\infty}^{\infty} Y_r X_{n-r} \\ &= \sum_{r=-\infty}^{\infty} X_{n-r}Y_r = X_n * Y_n \end{aligned}$$

[4]

Surprisingly many people found this difficult and did not realise that it just involved an index substitution. Some tried to use the convolution theorem (that $X_n * Y_n$ *is the Fourier series for* $x(t)y(t)$ *and* $x(t)y(t) = y(t)x(t)$*). This is an acceptable proof (albeit a bit of a cheat) but in fact almost everyone who used this method got it wrong. Some substituted* $n = m+r$ *of the left of the equation as well; this is entirely wrong - you must never use a summation dummy variable outside the sum. Quite a lot of people were a bit vague about what happens to the summation limits when you make the substitution* $m = n-r$*; the order of summation is actually reversed, but this leaves the sum unaffected provided that the sum converges absolutely (which we assume it does since we are told that* X_n *and* Y_n *are Fourier coefficients. One or two people changed the summation given in the question to an integral; this is completely wrong for several reasons one of which is that, since m is an integer, it cannot be an integration variable. Quite a few people asserted (without proof) that* $X_{n-m}Y_m = Y_{n-m}X_m$ *which is not generally true.*

i) Since $x(t)$ is real, $x^*(t) = x(t)$ so we can write $y(t) = x(t) \otimes x(t) = \int_{-\infty}^{\infty} x(s-t)x(s)ds$. We know that $x(s)$ is zero for $s < 0$ and hence $x(s-t)$ is zero for $s < t$. It follows that the integrand will be zero for $s < \max(0, t)$.

Thus we get two different lower integration limits depending on whether or not

$t < 0$. When $t < 0$

$$\begin{aligned} y(t) &= \int_0^\infty e^{-2(s-t)} e^{-2s} ds \\ &= e^{2t} \int_0^\infty e^{-4s} ds \\ &= -\frac{1}{4} e^{2t} \left[e^{-4s} \right]_0^\infty = \frac{1}{4} e^{2t} \end{aligned}$$

When $t \geq 0$ the integrand is the same but the lower limit is now t , so

$$\begin{aligned} y(t) &= -\frac{1}{4} e^{2t} \left[e^{-4s} \right]_t^\infty \\ &= \frac{1}{4} e^{2t} e^{-4t} = \frac{1}{4} e^{-2t} \end{aligned}$$

We can neatly combine these into $y(t) = \frac{1}{4} e^{-2|t|}$. [4]

Most people realized that the integrand was 0 for $s < 0$ but very few realized that it is also zero for $s < t$; thus many only obtained the first part of the answer given above. Many people said that $x(s-t) = e^{2(s-t)}$ rather than $e^{-2(s-t)}$. One or two wrongly said the conjugate of e^{-2t} was e^{-2it} whereas others said it was e^{+2t}. When evaluating the integral, a few people applied the integration limits to t rather than to s. Several said $x(s) = e^{-2t}$ because they failed to substitute the dummy variable s into the definition. In the question, the dummy integration valiable is given as s but several people changed it into t instead; this is not valid because then the dummy integration variable is used outside the integral in $y(t) = \cdots$ which is not allowed.

j) From the formula sheet,

$$\begin{aligned} x(t) &= \int_{-\infty}^\infty X(f) e^{i2\pi ft} df \\ &= 2i \left(\int_{-\infty}^\infty \delta(f+20) e^{i2\pi ft} df - \int_{-\infty}^\infty \delta(f-20) e^{i2\pi ft} df \right) \\ &= 2i \left(e^{-i40\pi t} - e^{i40\pi t} \right) \\ &= 2i \times -2i \sin 40\pi t \\ &= 4 \sin 40\pi t \end{aligned}$$

[4]

Mostly well done. A few people included a factor of $1/2\pi$ in the integral - this is only needed when integrating with respect to ω rather than f. Several people got the formula for the inverse Fourier transform wrong even though it was given in the formula sheet. A few people confused the Fourier Transform and the Fourier Series: the first involves a function of frequency, $X(f)$, whereas the second involves discrete coefficients, X_n. Note that $\delta(f+20)$ is an impulse at $f = -20$; it is easy to get the sign wrong which results in $x(t) = -4 \sin 40\pi t$. Several people mixed up the frequency and angular-frequency versions of the Fourier Transform; whereas both transforms give the same spectrum for aperiodic signals, the slightly peculiar properties of delta functions mean that $\cos(2\pi f_0 t) \to 0.5(\delta(f-f_0) + \delta(f+f_0))$ whereas $\cos(\omega_0 t) \to \pi(\delta(\omega-\omega_0) + \delta(\omega+\omega_0))$. This arises because $\delta(f) = 2\pi\delta(2\pi f)$; this equivalence is givien in the formula sheet.

2. Given the function

$$f(x) = \frac{x}{1+x^4} - \frac{x^3}{1+x^4} \tag{2.1}$$

a) Show that the area under the graph of the function $f(x)$, for $0 \le x \le b$ is given by

$$A(b) = \frac{1}{2}\arctan(b^2) - \frac{1}{4}\log(1+b^4)\,. \tag{2.2}$$

The integral is made of two parts, the first requires a trigonometric substitution, the second is an almost exact differential.
For the first integral:

$$\begin{aligned} A_1(b) &= \int_0^b \frac{x}{1+x^4}\,dx \\ &\text{substitute } x^2 = u \to 2x\,dx = du \\ A_1(b) &= \frac{1}{2}\int \frac{1}{1+u^2}\,du = \\ &= \frac{1}{2}\arctan(u) \\ &= \frac{1}{2}\left[\arctan(x^2)\right]_0^b = \\ &= \frac{1}{2}\arctan(b^2) \end{aligned}$$

[3]

For the second integral:

$$\begin{aligned} A_2(b) &= -\int_0^b \frac{x^3}{1+x^4}\,dx = \\ &-\frac{1}{4}\int_0^b \frac{4x^3}{1+x^4}\,dx = \\ &-\frac{1}{4}\left[\log(1+x^4)\right]_0^b = -\frac{1}{4}\log(1+b^4). \end{aligned}$$

[3]

The result follows.

For questions of the form "Show that XX is equal to YY" it is important to give enough detail to show that you have understood: some of you wrote down the solution, sometimes just fitting it into the given answer, which is not enough.

b) Find the stationary points of $A(b)$ with $b > 0$ and determine whether they are maxima or minima.

Must now calculate the first derivative of $A(b)$ and its roots, but the result (the first derivative) is given by the initial integrand and the calculation is superfluous.

$$\frac{dA}{db} = \frac{b-b^3}{1+b^4}.$$

Critical points are given by

$$b - b^3 = 0 \to b_c = 0, \pm 1 \to b_c = +1.$$

(as $b > 0$, only the positive root is accepted). [3]

A large number of people tried differentiating $A(b)$ instead of just taking $f(b)$, often resulting in errors. An even larger number of people ignored the restriction $b > 0$. Some didn't ignore it, and correctly rejected $b = -1$, but then asserted stationary points at $b = 0, 1$!

To check the nature of the stationary points, calculate the second derivative

$$\frac{d^2A}{db^2} = \frac{(1-3b^2)(1+b^4) - 4b^3(b-b^3)}{(1+b^4)^2},$$

which, at the critical point $b_c = +1$ gives

$$\frac{d^2A}{db^2} = -1 < 0$$

therefore b_c is a maximum.

[3]

Mostly fine. A few people tried to argue about the behaviour of $A'(b)$ changing sign near the stationary point, without calculating $A''(b)$, which is preferable.

c) Assume that b is a function of time given by

$$b(t) = e^{-t}. \tag{2.3}$$

i) Use the chain rule to determine $\dfrac{\mathrm{d}A}{\mathrm{d}t}$ as a function of t.

Using the chain rule and the results of the previous part of the question:

$$\frac{dA}{dt} = \frac{dA}{db}\frac{db}{dt} = \frac{b-b^3}{1+b^4}\left(-e^{-t}\right) = \frac{e^{-4t} - e^{-2t}}{e^{-4t}+1}.$$

[4]

Mostly fine. Many left this as $-\dfrac{b-b^3}{1+b^4}e^{-t}$. Not enough!

ii) Find the limit of $A(b(t))$ as t tends to $+\infty$.

$$\lim_{t\to\infty} b(t) = \lim_{t\to\infty} e^{-t} = 0, \text{ thus}$$
$$\lim_{t\to\infty} \frac{1}{2}\arctan(b^2) - \frac{1}{4}\log(1+b^4) =$$
$$\lim_{b\to 0} \frac{1}{2}\arctan(b^2) - \frac{1}{4}\log(1+b^4) = 0.$$

[4]

Mostly well done. A lot of people calculated the limit of $-\dfrac{b-b^3}{1+b^4}e^{-t}$ instead!

3. a) i) Show that $\sinh(x+iy) = \sinh x\cos y + i\cosh x\sin y$.

$$\begin{aligned}\sinh(x+iy) &= \frac{1}{2}\left(e^{x+iy} - e^{-x-iy}\right)\\ &= \frac{1}{2}\left[e^x(\cos y + i\sin y) - e^{-x}(\cos y - i\sin y)\right]\\ &= \cos y\left(\frac{e^x - e^{-x}}{2}\right) + i\sin y\left(\frac{e^x + e^{-x}}{2}\right)\\ &= \cos y\sinh x + i\sin y\cosh x\end{aligned}$$

as required. [4]

A large number of variations. A few noticed the simpler $\sinh(x+iy) = \sinh x\cosh(iy) + \cosh x\sinh(iy)$ *. But many tried much more complicated approaches, often based on converting* $\sinh(x+iy) = \sinh(i(y-ix)) = i\sin(y-ix)$*, or writing* $\cos y\sinh x + i\sin y\cosh x = \cosh(iy)\sinh x + \sinh(iy)\cosh x$ *and then writing out all the hyperbolic functions in terms of exponentials. The more complicated the approach, the more errors were made.*

ii) Hence, or otherwise, show that $|\sinh(x+iy)|^2 = \frac{1}{2}(\cosh 2x - \cos 2y)$.

From (i) we have

$$\begin{aligned}|\sinh(x+iy)|^2 &= \cos^2 y\sinh^2 x + \sin^2 y\cosh^2 x\\ &= (1-\sin^2 y)\sinh^2 x + \sin^2 y(1+\sinh^2 x)\\ &= \sinh^2 x + \sin^2 y\\ &= \frac{1}{2}(\cosh 2x - 1) + \frac{1}{2}(1 - \cos 2y)\end{aligned}$$

and the result follows. [4]

Many good results, with enormous variation of approach. Again, the more complicated versions resulted in more errors. For example, many reverted to exponentials or made changes like $\cosh(x) = \cos(ix)$*, etc. A number of people assumed* $|\sinh(x+iy)|^2 = (\cos y\sinh x + i\sin y\cosh x)^2$*, unfortunately.*

b) Obtain the limit: $\lim_{x\to -3}\dfrac{3-\sqrt{-3x}}{x+3}$. [Do NOT use l'Hopital's rule.]

$$\begin{aligned}\lim_{x\to -3}\frac{3-\sqrt{-3x}}{x+3} &= \lim_{x\to -3}\frac{(3-\sqrt{-3x})}{(x+3)}\frac{(3+\sqrt{-3x})}{(3+\sqrt{-3x})}\\ &= \lim_{x\to -3}\frac{9+3x}{(x+3)(3+\sqrt{-3x})}\\ &= \lim_{x\to -3}\frac{3}{3+\sqrt{-3x}}\\ &= \frac{1}{2}\end{aligned}$$

[4]

A large minority wrote $\dfrac{(3-\sqrt{-3x})}{(x+3)}\dfrac{(3+\sqrt{-3x})}{(3+\sqrt{-3x})} = \dfrac{9+3x}{3x+9+x\sqrt{-3x}+3\sqrt{-3x}}$ *and then failed to see the cancellation. Those that didn't multiply out the de-*

nominator usually got it right. A few people multiplied numerator and denominator by $x-3$.

c) Show that $\displaystyle\int \frac{dx}{\sqrt{x^2-1}} = \cosh^{-1} x + C$.

Let $x = \cosh u$, then $dx = \sinh u\, du$ and we get

$$\begin{aligned}\int \frac{dx}{\sqrt{1-x^2}} &= \int \frac{\sinh u\, du}{\sqrt{\cosh^2 u - 1}} \\ &= \int \frac{\sinh u\, du}{\sqrt{\sinh^2 u}} \\ &= \int 1\, du \\ &= u + C \\ &= \cosh^{-1} x + C\end{aligned}$$

[4]

Mostly fine. A few people tried an odd approach, writing $\sqrt{x^2-1} = i\sqrt{1-x^2}$ and then using a trigonometric substitution, unsuccessfully.

d) Integrate $\displaystyle\int \frac{1}{1-\sin x - \cos x}\, dx$.

Use the standard substitution from formula sheet with $t = \tan(x/2)$ then

$$\begin{aligned}\int \frac{1}{1-\sin x - \cos x}\, dx &= \int \frac{1}{1 - \frac{1-t^2}{1+t^2} - \frac{2t}{1+t^2}} \left(\frac{2\, dt}{1+t^2}\right) \\ &= \int \frac{2}{1+t^2-(1-t^2)-2t}\, dt \\ &= \int \frac{1}{t^2-t}\, dt \\ &= \int \frac{1}{t-1} - \frac{1}{t}\, dt \\ &= \ln|t-1| - \ln|t| + C \\ &= \ln\left|\tan\left(\frac{x}{2}\right) - 1\right| - \ln\left|\tan\left(\frac{x}{2}\right)\right| + C\end{aligned}$$

[4]

Most people did this well, with the odd arithmetic error slipping in. Many left the answer in terms of t. A few tried multiplying numerator and denominator by $1+\cos x + \sin x$ which leads to three more difficult integrals.

4. a) We know that $x(t) = \sum_{n=-\infty}^{\infty} X_n e^{i2\pi nFt}$. Hence

$$\begin{aligned}\frac{1}{T}\int_0^T |x(t)|^2 &= \langle x(t)x^*(t)\rangle \\ &= \left\langle \sum_{n=-\infty}^{\infty} X_n e^{i2\pi nFt} \sum_{m=-\infty}^{\infty} X_m^* e^{-i2\pi mFt} \right\rangle \\ &= \sum_{n=-\infty}^{\infty} X_n \sum_{m=-\infty}^{\infty} X_m^* \left\langle e^{i2\pi(n-m)Ft} \right\rangle\end{aligned}$$

But $\left\langle e^{i2\pi kFt}\right\rangle = \langle \cos 2\pi kFt\rangle + i\langle \sin 2\pi kFt\rangle = \begin{cases} 1 & \text{for } k = 0 \\ 0 & \text{for } k \neq 0 \end{cases}$

So all terms of $\sum_{m=-\infty}^{\infty} X_m^* \left\langle e^{i2\pi(m-n)Ft}\right\rangle$ are zero except for the one with $m = n$ which means that the entire sum collapses to $\sum_{m=-\infty}^{\infty} X_m^* \left\langle e^{i2\pi(m-n)Ft}\right\rangle = X_n^*$.

Hence

$$\frac{1}{T}\int_0^T |x(t)|^2 = \sum_{n=-\infty}^{\infty} X_n X_n^* = \sum_{n=-\infty}^{\infty} |X_n|^2$$

An alternative, rather neater, proof is obtained by substituting only for $x^*(t)$ in the first line:

$$\begin{aligned}\langle x(t)x^*(t)\rangle &= \left\langle x(t) \sum_{n=-\infty}^{\infty} X_n^* e^{-i2\pi nFt} \right\rangle \\ &= \left\langle \sum_{n=-\infty}^{\infty} X_n^* x(t) e^{-i2\pi nFt} \right\rangle \\ &= \sum_{n=-\infty}^{\infty} X_n^* \left\langle x(t) e^{-i2\pi nFt}\right\rangle \\ &= \sum_{n=-\infty}^{\infty} X_n^* X_n = \sum_{n=-\infty}^{\infty} |X_n|^2\end{aligned}$$

[5]

Many people found this difficult. Some people just stated (correctly) that this was Parseval's theorem from the formula sheet; this however does not constitutue a proof. The commonest mistake was to assume that you could interchange squaring and summation to give $\left|\sum_n X_n e^{i2\pi nFt}\right|^2 = \sum_n \left|X_n e^{i2\pi nFt}\right|^2$*; this is not generally true (e.g.* $(a+b)^2 \neq a^2 + b^2$*). Some people used the same dummy variable, n, in both summations writing* $\left|\sum_n V_n\right|^2 = \left(\sum_n V_n\right)\left(\sum_n V_n^*\right)$*; it is essential to use distinct dummy variables whenever summations are multiplied together. A few people wrote* $|x(t)|^2 = x(t) \times x(t)$ *rather than* $|x(t)|^2 = x(t) \times x^*(t)$.

b) For convenience, we will write $a = -i2\pi nF = -i\pi n$. Then

$$\begin{aligned}
X_n &= \frac{1}{2}\int_0^2 x(t)e^{at}dt \\
&= \frac{1}{2}\int_0^1 te^{at}dt + \frac{1}{2}\int_1^2 (2-t)e^{at}dt \\
&= \frac{1}{2}\left[\left(\frac{t}{a}-\frac{1}{a^2}\right)e^{at}\right]_0^1 + \frac{1}{2}\left[\left(\frac{2-t}{a}+\frac{1}{a^2}\right)e^{at}\right]_1^2 \\
&= \frac{1}{2}\left(\left(\frac{1}{a}-\frac{1}{a^2}\right)e^{a}+\frac{1}{a^2}\right) + \frac{1}{2}\left(\frac{1}{a^2}e^{2a}-\left(\frac{1}{a}+\frac{1}{a^2}\right)e^{a}\right) \\
&= \frac{1}{2}\left(\frac{1}{a^2}e^{2a}-\frac{2}{a^2}e^{a}+\frac{1}{a^2}\right) = \frac{1}{2a^2}\left(e^{2a}-2e^{a}+1\right)
\end{aligned}$$

Now we substitute for $a = -i\pi n$ and use $e^{i\pi} = -1$ to get

$$\begin{aligned}
X_n &= \frac{-1}{2\pi^2 n^2}\left(e^{-i2\pi n}-2e^{-i\pi n}+1\right) \\
&= \frac{-1}{2\pi^2 n^2}\left(1-2\left(-1\right)^n+1\right) \\
&= \frac{(-1)^n-1}{\pi^2 n^2}.
\end{aligned}$$

[6]

A simplifying alternative is to integrate instead from -1 to $+1$,

$$\begin{aligned}
X_n &= \frac{1}{2}\int_{-1}^1 x(t)e^{at}dt \\
&= \frac{1}{2}\int_{-1}^0 -te^{at}dt + \frac{1}{2}\int_0^1 te^{at}dt \\
&= \frac{1}{2}\int_1^0 \tau e^{-a\tau}(-d\tau) + \frac{1}{2}\int_0^1 te^{at}dt \\
&= \frac{1}{2}\int_0^1 \tau e^{-a\tau}d\tau + \frac{1}{2}\int_0^1 te^{at}dt \\
&= \frac{1}{2}\int_0^1 t(e^{at}+e^{-at})dt = \int_0^1 t\cos(-\pi nt)dt = \int_0^1 t\cos(\pi nt)dt \\
&= \left[\frac{1}{\pi n}t\sin(\pi nt)+\frac{1}{(\pi n)^2}\cos(\pi nt)\right]_0^1 \\
&= \frac{1}{\pi^2 n^2}(\cos(n\pi)-1) = \frac{(-1)^n-1}{\pi^2 n^2}.
\end{aligned}$$

Most people were able to evaluate the integral although sometimes with sign errors. Substituting $a = -i2\pi nF = -i\pi n$, as is done above, greatly reduces the mess and also the number of minus signs to keep track of; however very few people did this. Quite a few people omitted the i from the exponent for some reason. Since $x(t)$ is even, you can also say that X_n will equal twice the real part of the integral from 0 to 1 which directly gives $X_n = \Re\left(\int_0^1 te^{-i\pi nt}dt\right) = \int_0^1 t\cos(\pi nt)dt$; a few people took this approah but often omitted to take only the real part of the integral.

c) Note that the expression for X_n is even, so $X_{-n} = X_n$. Also the expression gives $X_0 = \frac{0}{0}$ so we need to calculate X_0 directly from the definition.

$X_0 = \langle x(t)e^{0t} \rangle = \langle x(t) \rangle = 0.5$ where this value is deduced from the average value of the waveform. Alternatively, an algebraic calculation gives

$$\begin{aligned} X_0 = \langle x(t) \rangle &= \frac{1}{2}\int_0^2 x(t)dt \\ &= \frac{1}{2}\left(\int_0^1 t\,dt + \int_1^2 (2-t)\,dt\right) \\ &= \frac{1}{2}\left(\left[\frac{t^2}{2}\right]_0^1 + \left[2t - \frac{t^2}{2}\right]_1^2\right) \\ &= \frac{1}{2}\left(\left(\frac{1}{2} - 0\right) + \left(2 - \frac{3}{2}\right)\right) \\ &= \frac{1}{2} = 0.5 \end{aligned}$$

For the other values of n, we can use the formula: $X_{\pm 1} = \frac{-2}{\pi^2}$, $X_{\pm 2} = 0$, $X_{\pm 3} = \frac{-2}{9\pi^2}$, $X_{\pm 4} = 0$. [4]

Almost everyone got all values correct except X_0. Many people said that $X_0 = 0$ because the numerator of the formula in part b) was zero. Others said that $X_0 = 0/0 = 1$; this is not generally true: $0/0$ can equal anything. A few used L'Hopital's rule to evaluate the formula $X_n = \frac{(-1)^n - 1}{\pi^2 n^2}$ at $n = 0$ by differentiating the numerator and denominator twice. This is unfortunately not valid because in deriving the formula, we made use of $e^{-i2\pi n} = 1$ which assumes that $n \equiv 0$. If you want to use L'Hopital's rule, you must apply it to $X_n = \frac{-1}{2\pi^2 n^2}\left(e^{-i2\pi n} - 2e^{-i\pi n} + 1\right)$ directly. Instead of writing $X_3 = \frac{-2}{9\pi^2}$, some said $X_3 = \frac{-2}{n^2\pi^2}$; this actually makes no sense since n appears only on the right hand side of the equation.

d) i) If $x(t)$ is even, then the X_n are also even, i.e. $X_n = X_{-n}$. If $x(t)$ is both even and real-valued, then the X_n are also even and real-valued. [2]

Most people got this right although there was some confusion between the consequences of $x(t)$ being real and those of $x(t)$ being even. There was also confusion between the Fourier series and Fourier transform: some people gave the solution in terms of $X(f)$ even though the question asked about X_n.

ii) We can deduce two things.

Firstly

$$\begin{aligned} x(t) + x(t+1) &= 1 \\ \Rightarrow \quad \langle x(t) \rangle + \langle x(t+1) \rangle &= 1 \\ \Rightarrow \quad \langle x(t) \rangle &= 0.5 \\ \Rightarrow \quad X_0 &= 0.5 \end{aligned}$$

where the third line follows because $\langle x(t) \rangle = \langle x(t+a) \rangle$ for any a.

Secondly we can write

$$\begin{aligned} (x(t) - 0.5) + (x(t+1) - 0.5) &= 0 \\ \Rightarrow \quad (x(t) - 0.5) &= -\left(x(t + \frac{T}{2}) - 0.5\right) \\ \Rightarrow \quad (x(t) - X_0) &= -\left(x(t + \frac{T}{2}) - X_0\right) \end{aligned}$$

which means that $(x(t) - X_0)$ is anti-periodic and so all its even numbered coefficients are zero. It follows that all the even numbered coefficients of $x(t)$ except for X_0 are zero.

An alternative derivation is to use the shift property of Fourier transforms. If $y(t) = x(t+1)$ then $Y_n = X_n e^{i2\pi nF\times 1} = -1^n X_n$. Also if $z(t) = 1$ then $Z_n = \delta_n$ which equals 0 except when $n = 0$ when it equals 1. Taking the Fourier transform of $x(t) + x(t+1) = 1$ or, equivalently, $x(t) + y(t) = z(t)$, then gives $X_n + -1^n X_n = \delta_n$ from which follows the two relationships given above. [3]

Many fewer go this right. Quite a lot of people said that $x(t)$ was antiperiodic which would mean $x(t)+x(t+1) = 0$ rather than 1. Most people who used the shift property, omitted to take the Fourier transform of the right-hand-side and often wrongly said $X_n + -1^n X_n = 1$